HOCHSCHULE
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OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 4: Voltage References

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Contents

4 Voltage References **3**

4.1 Why Voltage References? 3
4.2 Specifications 4
4.3 The Zener Reference 5
4.4 The Bandgap Reference 8
4.5 Shunt and Series References 11
4.6 Buffering and Scaling a Reference 13
4.7 Ratiometric Measurement 15
4.8 Error Budget 16

References **18**

4 Voltage References

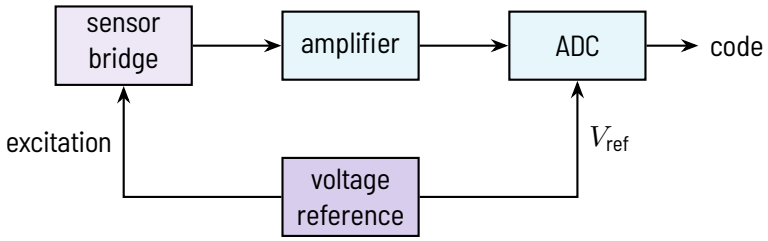
4.1 Why Voltage References?

Every measurement compares an unknown quantity with a known one. In an electronic measurement chain the known quantity is almost always a voltage: the *reference voltage*. An analog-to-digital converter does not measure volts, it measures the ratio of its input voltage to its reference voltage. A bridge circuit does not deliver millivolts per kelvin, it delivers a fraction of its excitation voltage. If the reference is $1\,\%$ too high, the result is $1\,\%$ wrong, no matter how good the rest of the circuit is.

A supply voltage from a voltage regulator is not good enough for this job. A typical regulator has an initial tolerance of $2\,\%$ … $5\,\%$, changes with load and temperature and carries ripple and noise from the digital part of the circuit. A voltage reference is a circuit (usually one small chip) that is optimised for a single purpose: a voltage that is accurate, stable over temperature and time, and quiet. It is usually not meant to deliver much current.

In this chapter we look at the specifications of references, at the two physical principles behind almost all of them (the Zener diode and the bandgap of silicon), at the shunt reference TL431 and at the practical questions: how to buffer and scale a reference, how to drive the reference ladder of a flash converter, and how a *ratiometric* design removes the reference error from the result altogether. The capstone task needs exactly this: a $4.00\,\mathrm{V}$ reference for the ADC ladder and for the sensor bridge [1, 6].

The reference sets the scale of the whole chain



- ADC: code $= 2^N \cdot V_{in}/V_{ref}$ (an ADC measures a *ratio*)
- bridge: output is a fraction of the excitation voltage
- an error of V_{ref} is a **gain error** of the complete chain

Regulator or reference?

	voltage regulator	voltage reference
purpose	supply power	define a voltage
initial accuracy	$1\,\%$ ... $5\,\%$	$0.02\,\%$ ... $1\,\%$
temperature coefficient	100 ppm/K ... 300 ppm/K	1 ppm/K ... 50 ppm/K
output current	mA ... A	µA ... 10 mA
noise	high	low, specified

- a reference is a "voltage standard" on the board, not a power supply
- if a load needs current: buffer the reference with an op-amp

4.2 Specifications

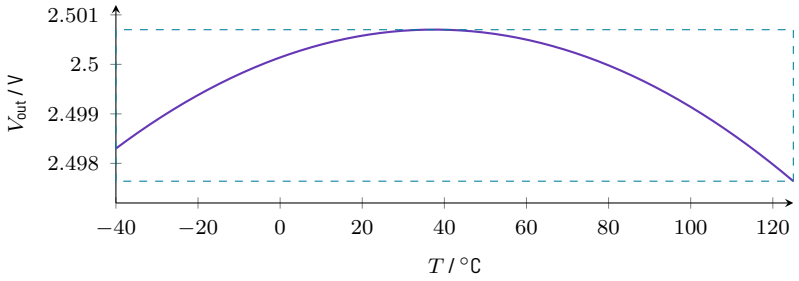
Data sheets of voltage references list a handful of parameters. They all describe how far the actual output voltage can be from its nominal value, but for different reasons: production spread, temperature, supply voltage, load current, noise and ageing. For an error budget we add them up (see the end of this chapter). The temperature coefficient is usually given in ppm/K (parts per million per kelvin): 1 ppm $= 10^{-6}$, so 100 ppm $= 0.01\,\%$.

Data sheet parameters

- **initial accuracy**: tolerance at $25\,°\mathrm{C}$, e.g. $\pm 0.1\,\%$
- **temperature coefficient** (TC): e.g. 20 ppm/K
- **line regulation**: $\Delta V_{\mathrm{out}}/\Delta V_{\mathrm{in}}$, e.g. $50\,\mu\mathrm{V/V}$
- **load regulation**: $\Delta V_{\mathrm{out}}/\Delta I_{\mathrm{out}}$, e.g. 100 ppm/mA
- **noise**: 0.1 Hz … 10 Hz peak-to-peak, e.g. $10\,\mu\mathrm{V}$ pp
- **long-term drift**: e.g. 50 ppm per 1000 h
- **thermal hysteresis**: change after a temperature cycle

The low-frequency noise from 0.1 Hz … 10 Hz is given separately because it cannot be filtered: it looks like a slow drift of the reference. Broadband noise above that can be reduced with a capacitor at a noise-reduction pin or with an RC filter after the reference.

Temperature coefficient: the box method



Box method

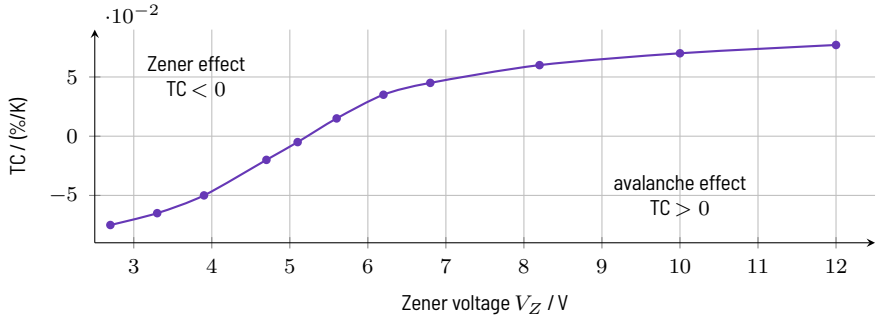
$$TC = \frac{V_{\mathrm{max}} - V_{\mathrm{min}}}{V_{\mathrm{nom}}\,(T_{\mathrm{max}} - T_{\mathrm{min}})} \cdot 10^{6}\ \mathrm{ppm}$$

The curve in the figure is typical for a bandgap reference: it is a parabola with its maximum near room temperature. The box method draws the smallest rectangle around the curve over the specified temperature range and divides its height by its width. In the example the box is 3.06 mV high and 165 K wide, so $TC = 3.06\,\mathrm{mV}/(2.5\,\mathrm{V} \cdot 165\,\mathrm{K}) \approx 7.4$ ppm/K. The TC does *not* mean that the voltage changes linearly; it only guarantees the box.

4.3 The Zener Reference

The oldest and simplest reference is a Zener diode operated in reverse breakdown. Above the breakdown voltage the current rises steeply, so the voltage stays nearly constant over a wide range of currents. Two physical effects cause the breakdown. Below about 5 V it is the *Zener effect* (quantum mechanical tunnelling through a thin, heavily doped junction); its breakdown voltage falls with temperature. Above about 6 V it is the *avalanche effect* (carriers gain enough energy to ionise further atoms); its breakdown voltage rises with temperature. Between 5 V and 6 V both effects are present and their temperature coefficients nearly cancel. This is why 5.6 V and 6.2 V Zener diodes are preferred as references [2].

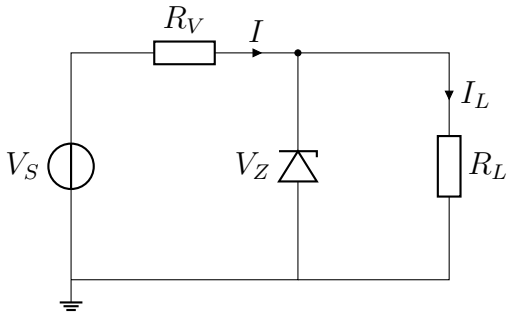
Temperature coefficient of Zener diodes



- minimum $|TC|$ near 5 V … 6 V: preferred reference Zeners 5.6 V, 6.2 V
- 6.2 V Zener ($+2$ mV/K) in series with a forward diode (-2 mV/K): compensated reference diode (1N821 family)

The curve shows typical values of small-signal Zener diodes; the spread between data sheets is large. The compensated reference diodes achieve 5 ppm/K … 100 ppm/K, but only at the specified current (typically 7.5 mA). The best references ever built (LM399, LTZ1000) use a *buried* Zener below the chip surface, which is far less noisy, together with an on-chip heater that keeps the die at a constant temperature.

Feeding a Zener through a resistor



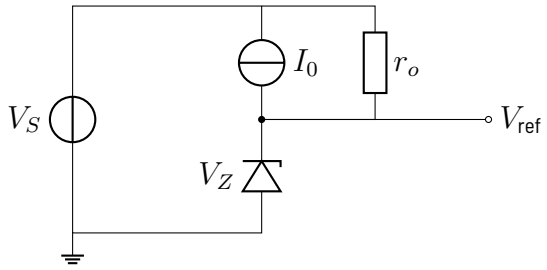
$$I_Z = I - I_L$$

- $R_V = \dfrac{V_S - V_Z}{I_Z + I_L}$, I_Z must stay above $I_{Z,\text{min}}$ for all V_S and I_L
- dynamic resistance $r_Z = \mathrm{d}V_Z/\mathrm{d}I_Z$: $5\,\Omega$ … $50\,\Omega$
- line regulation $\dfrac{\Delta V_Z}{\Delta V_S} \approx \dfrac{r_Z}{R_V + r_Z}$, load regulation $\Delta V_Z \approx -r_Z\,\Delta I_L$

▶ Falstad: `examples/references/zener_resistor.txt`

The Zener and the series resistor form a voltage divider whose lower part has the small dynamic resistance r_Z. A supply change is divided down by $r_Z/(R_V + r_Z)$, typically 1:100. The load current is taken from the Zener current: if the load draws more, the Zener carries less, and its voltage drops by $r_Z\,\Delta I_L$. Falstad models the Zener with an ideal exponential characteristic $I_Z = 5\,\text{mA}\cdot e^{(V-V_Z)/V_T}$, so the nominal voltage is reached at 5 mA and the dynamic resistance is only $V_T/I_Z \approx 5\,\Omega$. Real 5.6 V Zeners have $10\,\Omega$ … $40\,\Omega$, so in reality the regulation is several times worse than in the simulation.

Feeding a Zener from a current source



- the Zener current no longer depends on V_S
- only the output resistance r_o of a real current source remains: $\dfrac{\Delta V_Z}{\Delta V_S} \approx \dfrac{r_Z}{r_o}$
- e.g. transistor current source from chapter 3, $r_o = 100\,\text{k}\Omega$

▶ Falstad: `examples/references/zener_current_source.txt`

With $r_Z = 5\,\Omega$ and $r_o = 100\,\text{k}\Omega$ the line regulation improves from $1 : 240$ (with $R_V = 1.2\,\text{k}\Omega$) to $1 : 20000$. The same trick is used inside every integrated reference: the reference element is fed from a current that is derived from the reference voltage itself, so that supply changes do not reach it.

Exercise 4.1: Zener reference with series resistor

A Zener diode with $V_Z = 5.6\,\text{V}$ (at 5 mA) is fed from $V_S = 12\,\text{V}$ through $R_V = 1.2\,\text{k}\Omega$. Use the Falstad model of the Zener, $I_Z = 5\,\text{mA} \cdot e^{(V - V_Z)/V_T}$ with $V_T = 25.9\,\text{mV}$.

a) Calculate the Zener current and the dynamic resistance r_Z.

b) How much does V_{ref} change if V_S rises to $14\,\text{V}$?

c) A load $R_L = 2.2\,\text{k}\Omega$ is connected at $12\,\text{V}$. How much does V_{ref} drop?

Solution 4.1

a) $I_Z \approx (12\,\text{V} - 5.6\,\text{V})/1.2\,\text{k}\Omega = 5.33\,\text{mA}$, so $V_{\text{ref}} = V_Z + V_T \ln(5.33/5) = 5.602\,\text{V}$ and $r_Z = V_T/I_Z = 25.9\,\text{mV}/5.33\,\text{mA} = 4.9\,\Omega$.

b) At $14\,\text{V}$: $I_Z = 7.0\,\text{mA}$, $\Delta V_{\text{ref}} = V_T \ln(7.0/5.33) = 7.0\,\text{mV}$ (small-signal estimate with the mean $r_Z \approx 4.2\,\Omega$: $2\,\text{V} \cdot 4.2/1204 = 7.0\,\text{mV}$). Line regulation $3.5\,\text{mV/V}$, i.e. $0.06\,\%/\text{V}$.

c) The load takes $I_L = 5.6\,\text{V}/2.2\,\text{k}\Omega = 2.55\,\text{mA}$, the Zener keeps $I_Z = 5.33 - 2.55\,\text{mA} = 2.79\,\text{mA}$: $\Delta V_{\text{ref}} = V_T \ln(2.79/5.33) = -16.6\,\text{mV}$.

▶ Falstad: `examples/references/zener_resistor.txt`: the labels show `Vz12` $= 5.602\,\text{V}$, `Vz14` $= 5.609\,\text{V}$ ($+7.0\,\text{mV}$) and `Vz12L` $= 5.585\,\text{V}$ ($-16.6\,\text{mV}$).

Exercise 4.2: Zener with current-source feed

The Zener of the previous exercise is fed from a current source of $5\,\text{mA}$ with an output resistance $r_o = 100\,\text{k}\Omega$. How much does V_{ref} change when V_S rises from $12\,\text{V}$ to $14\,\text{V}$? Compare with the series resistor.

Solution 4.2

a) The current through r_o changes by $2\,\text{V}/100\,\text{k}\Omega = 20\,\mu\text{A}$, the Zener current from $5.064\,\text{mA}$ to $5.084\,\text{mA}$. $\Delta V_{\text{ref}} = V_T \ln(5.084/5.064) = 0.10\,\text{mV}$, 70 times less than with $R_V = 1.2\,\text{k}\Omega$ ($7.0\,\text{mV}$).

▶ Falstad: `examples/references/zener_current_source.txt`: at $12\,\text{V}$ `VzI` $= 5.600\,\text{V}$; moving the slider to $14\,\text{V}$ changes `VzR` by $7.0\,\text{mV}$ and `VzI` by only $0.1\,\text{mV}$.

4.4 The Bandgap Reference

Zener diodes need more than 6 V of supply and several milliamperes. Modern circuits run from 3.3 V or less and from microamperes. Almost all integrated references (and the references inside microcontroller ADCs) therefore use the *bandgap* principle, invented by Widlar [8] and brought to its classic form by Brokaw [9]. The idea: the base-emitter voltage of a transistor falls with temperature, and the difference of two base-emitter voltages at different current densities rises with temperature. A suitably weighted sum of the two does not depend on temperature. The sum turns out to be close to the bandgap voltage of silicon, 1.2 V, which gives the circuit its name.

Two voltages with opposite temperature coefficients

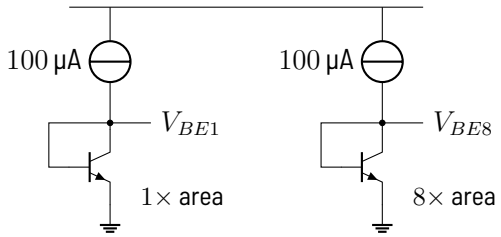
- base-emitter voltage: $V_{BE} = V_T \ln(I_C/I_S)$ with $V_T = kT/q = 25.7\,\text{mV}$ at $25\,°\text{C}$
  - I_S rises strongly with T: V_{BE} falls by about $2\,\text{mV/K}$ (CTAT: complementary to absolute temperature)
- two transistors with current densities in the ratio N:

Delta-VBE

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln N = \frac{kT}{q} \ln N$$ (PTAT: proportional to absolute temperature)

  - $k/q = 86.2\,\mu\text{V/K}$; $N = 8$: $\Delta V_{BE} = 53.5\,\text{mV}$ at $25\,°\text{C}$, $+0.179\,\text{mV/K}$
  - I_S cancels: ΔV_{BE} depends only on T and N

Delta-VBE in Falstad

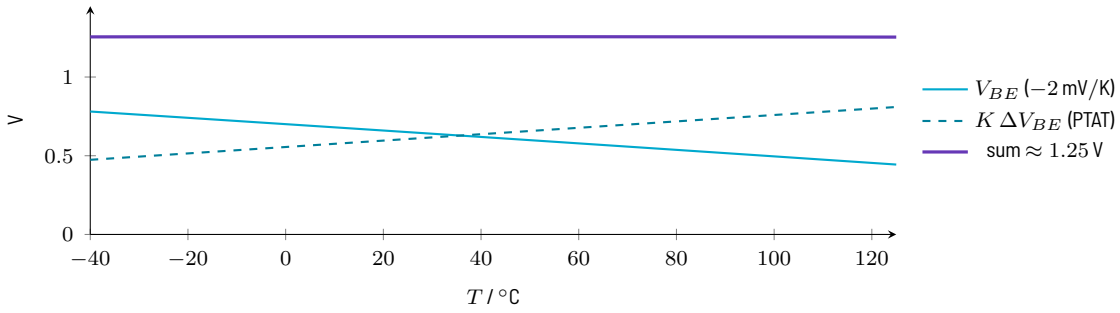


- same current, 8 transistors in parallel: current density $1/8$
- Falstad has no temperature, but ΔV_{BE} at room temperature can be measured: $V_{BE1} = 0.536\,\text{V}$, $V_{BE1} - V_{BE8} = 53.8\,\text{mV}$
- current ratio 10: $\Delta V_{BE} = V_T \ln 10 = 59.6\,\text{mV}$ ("60 mV per decade")

▶ Falstad: `examples/references/delta_vbe.txt`

Falstad uses $V_T = 25.865\,\text{mV}$ ($27\,°\text{C}$) and a transistor model with $I_S = 10^{-13}\,\text{A}$. The absolute value of V_{BE} is therefore lower than in a real small-signal transistor (about $0.65\,\text{V}$ at $100\,\mu\text{A}$), but the difference $V_T \ln N$ is exact, because I_S cancels. In an integrated circuit the area ratio is made by placing N identical transistors in parallel, exactly as in the simulation; this is the most precise ratio a chip can offer.

The bandgap principle

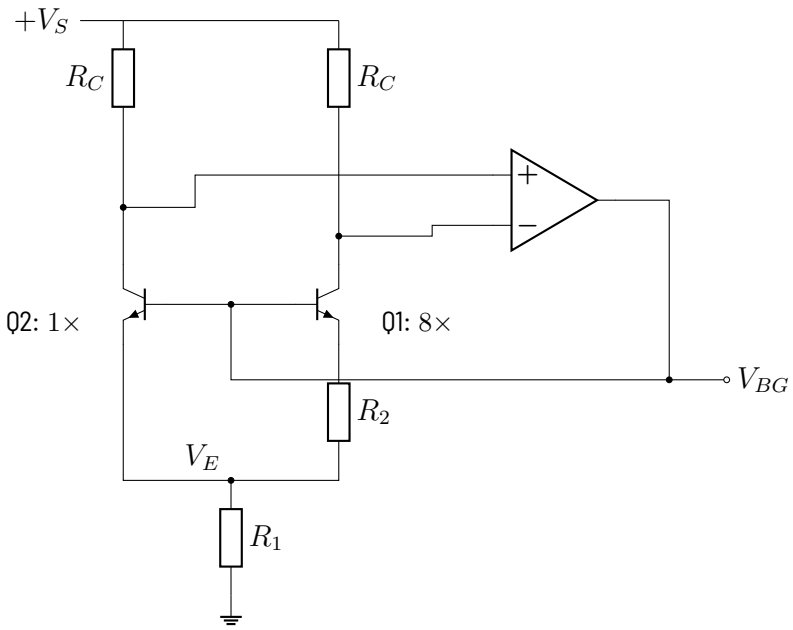


Bandgap reference

$V_{BG} = V_{BE} + K \cdot V_T \ln N \approx 1.25\,\text{V}$, K chosen so that $\mathrm{d}V_{BG}/\mathrm{d}T = 0$ at room temperature

The PTAT voltage must rise by $2\,\mathrm{mV/K}$. With $N = 8$ one ΔV_{BE} rises by $0.179\,\mathrm{mV/K}$, so it must be amplified by $K \approx 11$. At room temperature $K\,\Delta V_{BE} \approx 11 \cdot 54\,\mathrm{mV} \approx 0.6\,\mathrm{V}$, and the sum is $1.2\,\mathrm{V} \ldots 1.25\,\mathrm{V}$. Extrapolated to $0\,\mathrm{K}$, the sum is exactly the bandgap voltage of silicon, $V_{G0} = 1.205\,\mathrm{V}$. The remaining curvature (the $T \ln T$ term in V_{BE}) gives the parabola seen in the box-method figure; simple bandgaps reach $10\,\mathrm{ppm/K} \ldots 50\,\mathrm{ppm/K}$, curvature-corrected ones $1\,\mathrm{ppm/K} \ldots 5\,\mathrm{ppm/K}$.

The Brokaw cell



Q1 consists of eight transistors in parallel, Q2 of one. Both bases are driven by the output V_{BG}. The op-amp compares the voltages at the two equal collector resistors R_C and adjusts V_{BG} until both collector currents are equal. Then the current density in Q2 is eight times that in Q1, and the difference of the base-emitter voltages appears across R_2.

Brokaw cell: how it works

- the op-amp forces equal collector currents I (R_C equal, inputs at the collectors)
- current density ratio $N = 8$: $\Delta V_{BE} = V_T \ln 8$ appears across R_2
  - $I = V_T \ln 8 / R_2$ (PTAT current)
- R_1 carries both emitter currents: $V_E = 2 I R_1 = 2\,\dfrac{R_1}{R_2}\,V_T \ln 8$
- output: $V_{BG} = V_{BE2} + 2\dfrac{R_1}{R_2}\,V_T \ln 8$ ($K = 2R_1/R_2$)
- Falstad, $R_2 = 1\,\mathrm{k\Omega}$, $R_1 = 5.6\,\mathrm{k\Omega}$: $I = 53.8\,\mu\mathrm{A}$, $V_E = 0.602\,\mathrm{V}$, $V_{BE2} = 0.520\,\mathrm{V}$, $V_{BG} = 1.122\,\mathrm{V}$

▶ Falstad: `examples/references/brokaw_cell.txt`

With $K = 2 \cdot 5.6 = 11.2$ the PTAT part rises by $11.2 \cdot 0.179\,\mathrm{mV/K} = 2.0\,\mathrm{mV/K}$ and compensates V_{BE}. With a real transistor ($V_{BE} \approx 0.65\,\mathrm{V}$) the output would be $1.25\,\mathrm{V}$. Falstad's transistor model has a lower V_{BE} and no temperature dependence, so it shows $1.122\,\mathrm{V}$; move the slider of R_1 to see how the PTAT part scales ($R_1 = 6.8\,\mathrm{k\Omega}$ gives $1.25\,\mathrm{V}$). In a real chip R_1 is trimmed at the wafer test until the output has zero TC. A resistive divider between V_{BG} and the op-amp output (not shown) gives larger outputs such as $2.5\,\mathrm{V}$ or $4.096\,\mathrm{V}$.

Exercise 4.3: Bandgap design

A Brokaw cell has an area ratio $N = 8$, $R_2 = 1\,\text{k}\Omega$ and $R_1 = 5.6\,\text{k}\Omega$. Use $V_T = 25.865\,\text{mV}$ (as Falstad).

a) Calculate ΔV_{BE}, the collector current I and V_E.

b) The Falstad transistor model gives $V_{BE2} = 0.520\,\text{V}$ at this current. Calculate V_{BG}.

c) Calculate the temperature coefficient of the PTAT part and compare it with $\text{d}V_{BE}/\text{d}T \approx -2\,\text{mV/K}$.

Solution 4.3

a) $\Delta V_{BE} = 25.865\,\text{mV} \cdot \ln 8 = 53.8\,\text{mV}$, $I = 53.8\,\text{mV}/1\,\text{k}\Omega = 53.8\,\mu\text{A}$, $V_E = 2 \cdot 53.8\,\mu\text{A} \cdot 5.6\,\text{k}\Omega = 0.602\,\text{V}$.

b) $V_{BG} = 0.520\,\text{V} + 0.602\,\text{V} = 1.122\,\text{V}$.

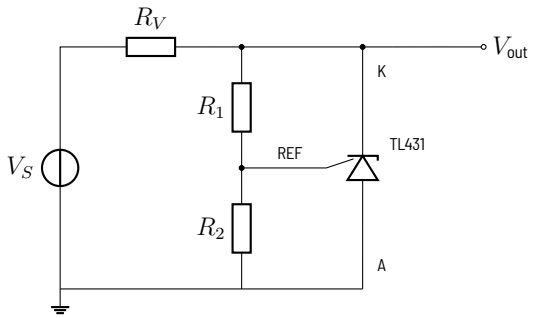
c) $\text{d}V_E/\text{d}T = 2\frac{R_1}{R_2}\frac{k}{q}\ln 8 = 11.2 \cdot 86.2\,\mu\text{V/K} \cdot 2.08 = +2.0\,\text{mV/K}$: it cancels the TC of V_{BE}.

▶ Falstad: `examples/references/brokaw_cell.txt`: VBG $= 1.122\,\text{V}$, VE $= 0.602\,\text{V}$; C1 and C2 are equal ($4.47\,\text{V}$), the voltage across R_2 is $53.8\,\text{mV}$.

4.5 Shunt and Series References

Integrated references come in two kinds. A *shunt* reference behaves like a precise Zener diode: it is connected in parallel to the load and draws exactly the current needed to keep its voltage constant; it is fed through a series resistor. A *series* reference behaves like a precise voltage regulator: it has an input, a ground and an output pin, and it only draws a small quiescent current itself. The most widely used shunt reference is the TL431, an adjustable bandgap reference with an internal $2.495\,\text{V}$ reference, an error amplifier and a shunt transistor.

The adjustable shunt reference TL431



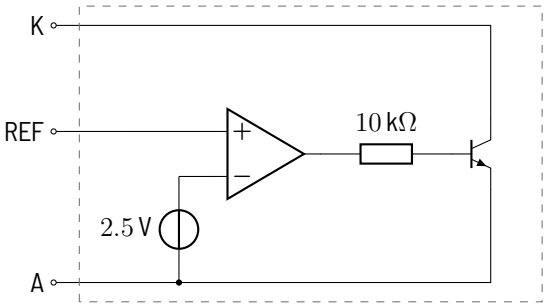
- the shunt element conducts until $V_{REF} = 2.5\,\text{V}$

TL431

$$V_{out} = V_{REF}\left(1 + \frac{R_1}{R_2}\right) + I_{REF}\,R_1, \quad V_{REF} = 2.495\,\text{V},\ I_{REF} \approx 2\,\mu\text{A}$$

- cathode current between 1 mA and 100 mA

Inside the TL431 (behavioural model)



- error amplifier compares REF with the internal 2.5 V bandgap
- V_{REF} too high ⇒ transistor conducts more ⇒ V_{out} falls
- $R_1 = 12\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$: $V_{out} = 2.5 \cdot 1.6 = 4.00\,\text{V}$

▶ Falstad: `examples/references/shunt_ref.txt`

The Falstad model sets the TL431 to 4.00 V, fed from 12 V through $1\,\text{k}\Omega$. The cathode current is $(12\,\text{V} - 4\,\text{V})/1\,\text{k}\Omega - 4\,\text{V}/32\,\text{k}\Omega = 7.875$ mA. Move the supply slider down to 6 V: the output stays at 4.00 V, only the cathode current falls. The internal reference is modelled as an ideal 2.5 V source; the real chip uses a bandgap cell.

Shunt or series reference?

- **shunt** (TL431, LM4040, LM385)
  - two terminals, works at any supply above V_{out}, can be used as a negative reference
  - the series resistor wastes current: R_V must deliver $I_{min} + I_{L,max}$ at the lowest supply
- **series** (REF50xx, ADR4xx, MAX6xxx)
  - three terminals, quiescent current only 0.1 mA … 1 mA
  - better for varying loads and battery operation, needs $V_{in} > V_{out} + V_{dropout}$
- both: decouple according to the data sheet (some are unstable with the wrong capacitor)

Exercise 4.4: TL431 set to 4.00 V

A TL431 ($V_{\text{REF}} = 2.495$ V, $I_{\text{REF}} = 2$ µA, $I_{K,\text{min}} = 1$ mA) shall provide 4.00 V from a supply of 6 V … 15 V.

a) Choose R_1 and R_2 from the E24 series.

b) How large is the error caused by I_{REF} and by $V_{\text{REF}} = 2.495$ V instead of 2.5 V?

c) Choose R_V such that a load of 1 mA can be supplied at 6 V.

Solution 4.4

a) $1 + R_1/R_2 = 1.6 \Rightarrow R_1/R_2 = 0.6$: $R_1 = 12$ kΩ, $R_2 = 20$ kΩ (both E24).

b) $I_{\text{REF}} R_1 = 2$ µA·12 kΩ $= 24$ mV ($+0.6$ %); 2.495 V·$1.6 = 3.992$ V (-0.2 %). Together $V_{\text{out}} = 4.016$ V: for an accurate 4.00 V choose smaller divider resistors or trim.

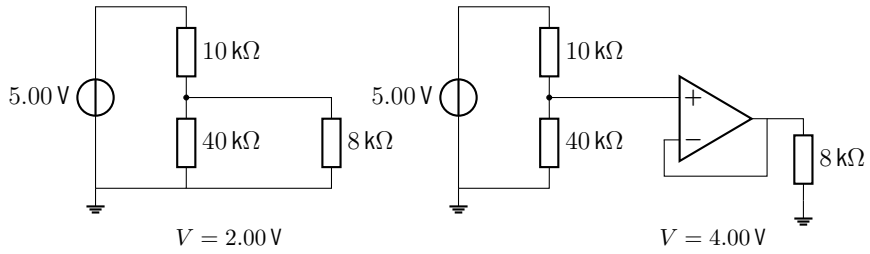
c) At 6 V R_V must carry $I_{K,\text{min}} + I_L + I_{\text{div}} = 1 + 1 + 0.125 = 2.125$ mA: $R_V \le (6\,\text{V} - 4\,\text{V})/2.125$ mA $= 941\ \Omega$, choose $910\ \Omega$. At 15 V the TL431 then carries $(15 - 4)/0.91 - 0.125 - 1 = 11$ mA (44 mW): fine.

▶ Falstad: `examples/references/shunt_ref.txt` (ideal model with 2.5 V, $R_V = 1$ kΩ, no load): `VK` $= 4.00$ V at 12 V and at 6 V, `REF` $= 2.50$ V.

4.6 Buffering and Scaling a Reference

References are available in a few standard voltages: 1.25, 2.048, 2.5, 3.0, 4.096 and 5.0 V. If the circuit needs a different value, or more current than the reference can deliver, an op-amp buffers and scales it. The capstone needs exactly 4.00 V for a reference ladder of eight 1 kΩ resistors, so that one LSB of the 3-bit flash converter is 0.5 V. The ladder draws $4\,\text{V}/8\,\text{k}\Omega = 0.5$ mA: far too much for a voltage divider with high-value resistors, and a noticeable load for many references. The op-amp circuits are treated in detail in chapter 6; here we only use the results.

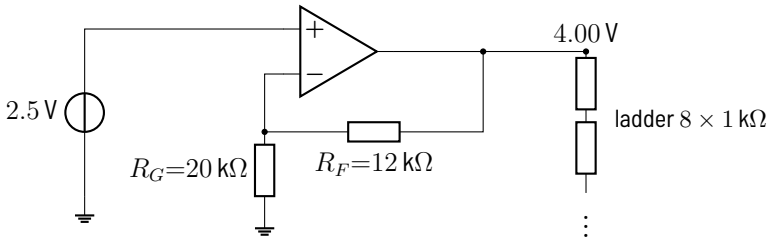
A divider is no reference



- divider $10\,\text{k}\Omega/40\,\text{k}\Omega$: 4.00 V open, but only 2.00 V with the $8\,\text{k}\Omega$ ladder
- source resistance $R_1 \parallel R_2 = 8\,\text{k}\Omega$ = load resistance
- follower: input current ≈ 0, output resistance ≈ 0: 4.00 V

▶ Falstad: `examples/references/ref_divider_loaded.txt`

Scaling with a non-inverting amplifier



- $G = 1 + R_F/R_G = 1 + 12/20 = 1.6$: $\quad 2.5\,\text{V} \cdot 1.6 = 4.00\,\text{V}$
- the op-amp delivers ladder current 0.5 mA plus 0.125 mA for $R_F + R_G$
- gain error from resistor tolerance: $\frac{\Delta G}{G} = \frac{G-1}{G}\left(\frac{\Delta R_F}{R_F} - \frac{\Delta R_G}{R_G}\right)$

▶ Falstad: `examples/references/ref_buffer_ladder.txt`

The resistor tolerances enter the result only with the factor $(G-1)/G = 0.375$: two $0.1\,\%$ resistors give at most $0.375 \cdot 0.2\,\% = 0.075\,\%$. The offset voltage of the op-amp is multiplied by G: 1 mV offset gives 1.6 mV at the output ($0.04\,\%$). Alternatively a 4.096 V reference with a follower can be used; the LSB is then 0.512 V instead of 0.5 V.

Design card

Design card: Reference for an ADC ladder

- full scale V_{FS} and LSB: $\text{LSB} = V_{FS}/2^N$ $\quad$ (4.00 V, 3 bit: 0.5 V)
- ladder current $I = V_{FS}/(2^N R)$ $\quad$ ($8 \times 1\,\text{k}\Omega$: 0.5 mA)
- reference voltage V_R from a standard value, gain $G = V_{FS}/V_R$ $\quad$ (2.5 V, $G = 1.6$)
- non-inverting amplifier: $R_F/R_G = G - 1$, $R_F + R_G \gg$ ladder ($12\,\text{k}\Omega + 20\,\text{k}\Omega$)
- check: op-amp output current, $(G-1)/G \cdot$ resistor tolerance, $G \cdot V_{\text{offset}}$
- decouple the ladder top with 100 nF … 1000 nF close to the comparators

Exercise 4.5: Reference for the 3-bit flash converter

The flash converter of the capstone needs 4.00 V at the top of a ladder of $8 \times 1\,\text{k}\Omega$. A 2.5 V reference is available.

a) Calculate the ladder current and the tap voltages.

b) Design a non-inverting amplifier with E24 resistors.

c) Why can the $4.00\,\text{V}$ not be taken from a divider $10\,\text{k}\Omega/40\,\text{k}\Omega$ at a $5\,\text{V}$ reference?

d) How large is the worst-case error of V_{ref} with $1\,\%$ resistors?

Solution 4.5

a) $I = 4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; taps every $0.5\,\text{V}$: $T_1 = 0.5\,\text{V}$, ..., $T_4 = 2.0\,\text{V}$, ..., $T_7 = 3.5\,\text{V}$.

b) $G = 4/2.5 = 1.6$, $R_F/R_G = 0.6$: $R_F = 12\,\text{k}\Omega$, $R_G = 20\,\text{k}\Omega$.

c) The divider has a source resistance of $10\,\text{k}\Omega \parallel 40\,\text{k}\Omega = 8\,\text{k}\Omega$, equal to the ladder. The loaded voltage is $5 \cdot \frac{40 \parallel 8}{10 + 40 \parallel 8} = 5 \cdot \frac{6.67}{16.67} = 2.00\,\text{V}$.

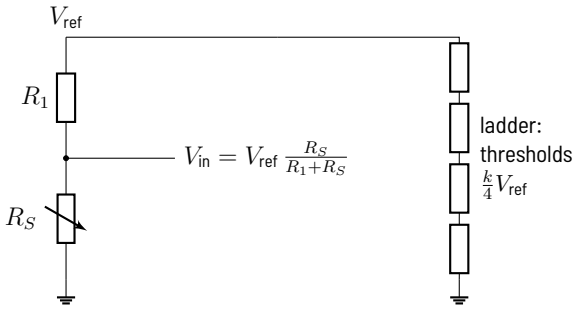
d) $\Delta G/G = 0.375 \cdot 2 \cdot 1\,\% = 0.75\,\%$, i.e. $\pm 30\,\text{mV}$; for the 3-bit converter (LSB $0.5\,\text{V}$) irrelevant, for a 12-bit converter 30 LSB.

▶ Falstad: `examples/references/ref_buffer_ladder.txt`: `VREF4` $= 4.00\,\text{V}$, `T4` $= 2.00\,\text{V}$, `T1` $= 0.50\,\text{V}$, ladder current $0.5\,\text{mA}$; ▶ Falstad: `examples/references/ref_divider_loaded.txt`: `Vopen` $= 4.00\,\text{V}$, `Vload` $= 2.00\,\text{V}$, `Vbuf` $= 4.00\,\text{V}$.

4.7 Ratiometric Measurement

Resistive sensors do not produce a voltage themselves; they need an excitation, and their output is a fraction of it. If the same reference excites the sensor and sets the full scale of the ADC, the reference appears in the numerator and in the denominator of the result and cancels. Such a measurement is called *ratiometric*. It is the most effective way to get an accurate result from an inexpensive reference: the reference only needs to be quiet and stable during one conversion, not accurate. The capstone uses this principle: the $4.00\,\text{V}$ reference feeds both the sensor bridge and the ladder of the flash converter [6].

The reference cancels



Ratiometric

code $= 2^N \dfrac{V_{\text{in}}}{V_{\text{ref}}} = 2^N \dfrac{R_S}{R_1 + R_S}$ independent of V_{ref}

Ratiometric versus absolute

- Falstad: two 2-bit flash converters (3 comparators), sensor divider $1\,\text{k}\Omega/2.7\,\text{k}\Omega$: $V_{\text{in}} = 0.730\,V_{\text{exc}}$
- left: sensor and ladder from V_{ref}: code 2 for $V_{\text{ref}} = 3.6, 4.0, 4.4$ V
- right: sensor from a fixed 4.00 V, ladder from V_{ref}
  - $V_{\text{ref}} = 4.0$ V: $V_{\text{in}} = 2.92$ V < 3.0 V: code 2
  - $V_{\text{ref}} = 3.6$ V: threshold $\frac{3}{4} V_{\text{ref}} = 2.7$ V: code 3 (wrong)
- ratiometric works only if both see the *same* voltage at the *same* time
  - a common connection with voltage drops, or noise on one path only, breaks it

▶ Falstad: `examples/references/ratiometric.txt`

4.8 Error Budget

An error budget lists all contributions to the error of a quantity and adds them up. Independent random errors are usually added as the root of the sum of squares (RSS), which is the realistic expectation; the simple sum is the worst case, which a guaranteed specification must meet. We look at the 4.00 V reference of the capstone, built from a 2.5 V series reference and the non-inverting amplifier, over an ambient temperature of $0\,^\circ$C … $50\,^\circ$C.

Error budget of the 4.00 V reference

contribution	data	error
initial accuracy	$\pm 0.2\,\%$	$0.20\,\%$
temperature	25 ppm/K, ± 25 K	$0.06\,\%$
gain resistors	$2 \times 0.1\,\%$, factor 0.375	$0.08\,\%$
op-amp offset	1 mV $\times$ 1.6	$0.04\,\%$
long-term drift	50 ppm / 1000 h	$0.01\,\%$
worst case (sum)		$0.39\,\%$
RSS		$0.23\,\%$

- compare: 3 bit, LSB $= 12.5\,\%$ of full scale: no problem at all
- 12 bit, LSB $= 0.024\,\%$: 16 LSB worst case $\Rightarrow$ calibrate or measure ratiometrically

The initial accuracy dominates. It can be removed by a one-time calibration (trimming, or a correction factor in the software). What remains after calibration is temperature drift and ageing, and these are the parameters that distinguish an expensive reference from a cheap one. In a ratiometric design all reference terms drop out; only the ratio accuracy of the resistors remains.

Exercise 4.6: Ratiometric measurement and error budget

A sensor divider $R_1 = 1\,\text{k}\Omega$, $R_S = 2.7\,\text{k}\Omega$ is read by a 2-bit flash converter with a ladder of $4 \times 1\,\text{k}\Omega$.

a) The divider and the ladder are fed from the same V_{ref}. Which code results for $V_{\text{ref}} = 4.0$ V and 3.6 V?

b) The divider is fed from a fixed 4.00 V, the ladder from V_{ref}. Repeat a).

c) A 12-bit converter uses a reference with $\pm 0.2\,\%$ initial accuracy and 25 ppm/K. How many LSB can the result be off at $50\,^{\circ}$C after calibration at $25\,^{\circ}$C?

Solution 4.6

a) $V_{\text{in}}/V_{\text{ref}} = 2.7/3.7 = 0.730$; the thresholds are at 0.25, 0.5 and $0.75\,V_{\text{ref}}$. $0.5 < 0.730 < 0.75$: code 2 for every V_{ref}.

b) $V_{\text{in}} = 2.92$ V. $V_{\text{ref}} = 4.0$ V: thresholds 1, 2, 3 V: code 2. $V_{\text{ref}} = 3.6$ V: thresholds 0.9, 1.8, 2.7 V: code 3. A $10\,\%$ reference error changes the result by one code.

c) After calibration only the drift remains: 25 ppm/K $\cdot\, 25$ K $= 625$ ppm $= 0.000625 \cdot 4096 = 2.6$ LSB. Without calibration add $0.002 \cdot 4096 = 8.2$ LSB.

▶ Falstad: `examples/references/ratiometric.txt`: with the VREF slider at 4.0 V both converters show D = E = code 2 (VIN1 $= 2.92$ V); at 3.6 V the right converter switches E3 on (code 3), the left stays at code 2.

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